Musanze Harvest Intelligence Service - Technical Report

INES Ruhengeri | SWE 3513 Artificial Intelligence | Individual Assignment 2

Item	Value
Student	Lazarus Simboya Ira Inyasio
Registration number	25/28180
Institutional username	ug2528180
Verification code	AI-A2-8180
Seed (LLLL)	8180
Dataset SHA-256	6e4324acd04d7c209b33860dc6eded0f 80048de0657a488bf124f65067644aae

1. Identity and decision problem

Stakeholder. A field officer at a potato cooperative in Musanze who receives farm records in inconsistent spreadsheets and needs a local, offline-capable tool.

Prediction questions. (1) Regression: how many tons of potatoes will a farm harvest? (2) Classification: is the farm at *high production risk* (yield per hectare below the median)? (3) Clustering: which farms share similar growing conditions?

Ethical boundary. Classroom decision-support prototype trained on synthetic data. It does not replace an agronomist and must not decide access to finance, insurance or public services. Human review is required. Predictions are advice for a human officer to review, never an automatic decision about a farmer.

Outputs. Command-line tool (`predict.py`) and a FastAPI service (`/health`, `/model-info`, `/predict`) for a future mobile app.

2. Data and reproducibility

The seed is the last four digits of the registration number (8180); `numpy.random.default_rng(8180)` drives every random step. Base rows = 240 + (8180 mod 41) = **261**. The generator then set m = round(0.05 x 261) = **13** values missing in each of two columns and appended three exact duplicate rows.

Column	Rule / role
rainfall_mm	Uniform 600-1400, feature
avg_temp_c	Uniform 12-26, feature
soil_moisture_pct	Uniform 20-80, feature (missing injected)
fertilizer_kg	Uniform 0-250, feature (missing injected)
field_size_ha	Uniform 0.2-4.0, feature
season	A or B, categorical feature
cooperative_support	0 or 1, binary feature
yield_tons	Formula + N(0, 0.80) noise, clipped at 0.5; regression target
high_risk	Yield per hectare below median; classification target

Data-quality evidence

Check	Result
Raw rows	264 (= 261 base + 3 duplicates)
Missing values (raw file)	soil_moisture_pct: 13, fertilizer_kg: 13
Exact duplicates detected	3
Duplicates removed	3
Clean rows used for modelling	261
Range / schema check	passed
Leakage check	passed

The raw file shows 13 missing values per column rather than 12 because one of the three duplicated rows copied a row that already had a missing value; removing duplicates restores 12 per column.

SHA-256 fingerprint of the CSV bytes (artifacts/data_fingerprint.txt)

`6e4324acd04d7c209b33860dc6eded0f80048de0657a488bf124f65067644aae`

Running `run_all.py` twice reproduces the same fingerprint because the CSV is written with four decimals and Unix line endings from the seeded generator; a test checks this.

3. Pipeline design

Split. One stratified 80/20 split with `random_state=8180`: 208 training and 53 test rows. The test set is used only for final reporting.

Preprocessing (one ColumnTransformer, fitted on training rows only). Numeric features: median imputation then standard scaling. Season: one-hot encoding. cooperative_support: passed through. The same transformer lives inside every saved model Pipeline, so `predict.py` and the API repeat exactly the training transformations.

Leakage controls. yield_tons, high_risk and yield per hectare are never features (`check_no_leakage` stops the run otherwise). Duplicates were removed before the split so a farm cannot appear in both train and test.

Candidates and selection. Each task has a baseline and three candidates. Selection uses 5-fold cross-validation inside the training set (RMSE for regression, F1 for classification); test scores are computed afterwards and not used to choose.

Task	Baseline	Candidates	Selected
Regression	mean	linear regression, random forest, gradient boosting	linear_regression
Classification	majority class	logistic regression, random forest, gradient boosting	logistic_regression
Clustering	-	KMeans, K=2..6, silhouette	K=6

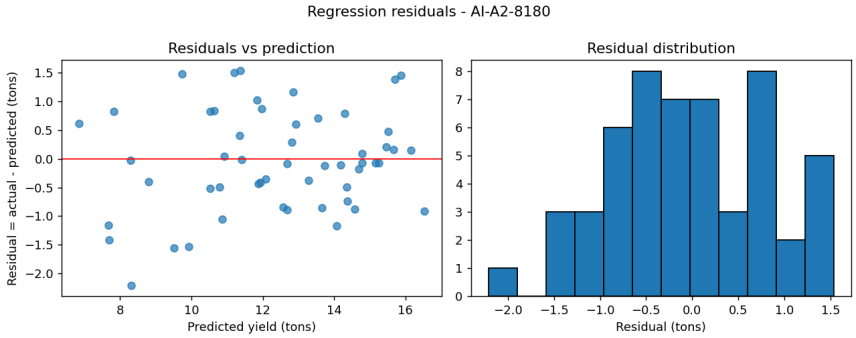
Why a simple model can win. The target formula is almost linear in the inputs, so linear and logistic models are strong candidates; tree ensembles need more data to match them with only 208 training rows.

Determinism. Every estimator, the split, the cross-validation folds and KMeans use random_state=8180. Versions: scikit-learn 1.8.0, Python 3.12.3.

4. Regression and classification (held-out test set, * = selected)

Regression

Model	MAE	RMSE	R2	CV RMSE (train)
mean_baseline	2.2937	2.8103	-0.0212	2.5075
linear_regression *	0.6962	0.8693	0.9023	1.0015
random_forest	1.1245	1.3634	0.7596	1.3337
gradient_boosting	0.9804	1.1970	0.8147	1.3076

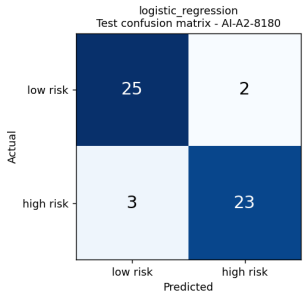


Residuals / selection. linear_regression had the lowest 5-fold cross-validated RMSE on the training set (1.0015 tons) and beats the mean baseline (test RMSE 2.8103 vs 0.8693). Selection used training-set CV only; test scores were reported afterwards.

Important error (row 172). Actual 6.10 t, predicted 8.31 t (error +2.21 t, 2.8 noise standard deviations). Because the data contains random noise with sd 0.80 t, the best possible RMSE is near 0.8 t; errors this size are mostly irreducible noise, not a model fault.

Classification (positive class = high risk)

Model	Precision	Recall	F1	ROC-AUC	CV F1 (train)
majority_baseline	0.000	0.000	0.000	0.500	0.131
logistic_regression *	0.920	0.885	0.902	0.972	0.929
random_forest	0.846	0.846	0.846	0.947	0.893
gradient_boosting	0.880	0.846	0.863	0.934	0.882



Confusion matrix (logistic_regression). TN=25, FP=2, FN=3, TP=23. logistic_regression had the highest 5-fold cross-validated F1 on the training set (0.9290). The majority baseline has test F1 0.0000 and recall 0.0000. Selection used training CV only; test scores were reported afterwards.

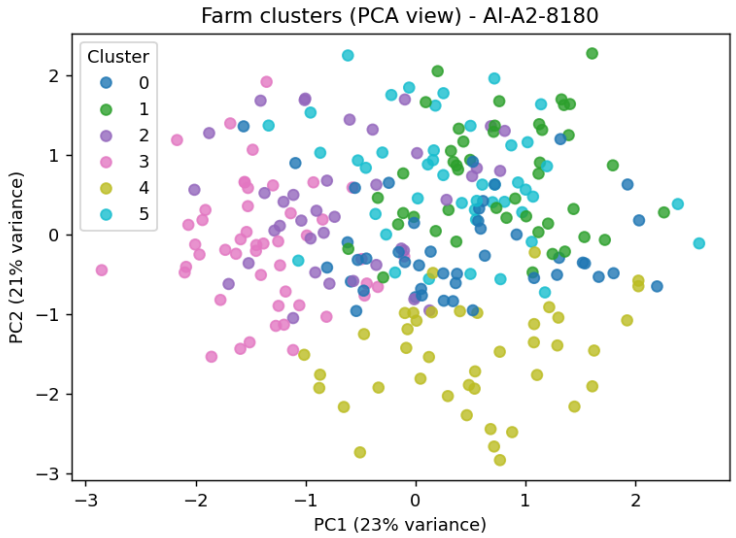
Important error (row 204). Actual class 0, predicted 1 (probability 0.77). Its yield per hectare was 5.64 versus a threshold of 5.33, so it lies close to the median cut-off where noise decides the label.

5. Clustering

KMeans was fitted on the five scaled numeric conditions of the training rows (targets unused). Season and cooperative support were left out of the distance because 0/1 columns made the first version simply split farms by season; they are used to describe clusters instead.

K	2	3	4	5	6
Silhouette	0.1653	0.1705	0.1817	0.1869	0.1943

Selected K = 6 (highest silhouette, 0.194). K=6 has the highest silhouette (0.194), which indicates weak cluster structure. Input features were generated independently, so clusters are rough groupings of similar farm conditions, not natural farm types. Targets were not used to fit clusters.



Cluster	Rows	Rain	Temp	Soil m.	Fert.	Size ha	Profile
0	32	1031	19.8	39	63	1.3	lower fertilizer use, lower field size, lower soil moisture, mixed seasons, mixed cooperative support.
1	33	1219	17.8	64	115	3.0	higher rainfall, higher soil moisture, higher field size, mixed seasons, mixed cooperative support.
2	29	947	22.1	65	172	1.2	higher soil moisture, lower field size, higher temperature, mixed seasons, mixed cooperative support.
3	43	735	14.9	56	144	2.2	lower rainfall, lower temperature, higher soil moisture, mixed seasons, mixed cooperative support.
4	31	1131	16.0	31	190	1.9	lower soil moisture, higher fertilizer use, higher rainfall, mixed seasons, mixed cooperative support.
5	40	885	22.2	42	96	3.2	higher field size, higher temperature, lower fertilizer use, mixed seasons, mixed cooperative support.

Operational meaning. A field officer can use the cluster label to compare a farm with others that have similar rainfall, soil moisture, fertilizer and field size, for example to plan advice visits. Because silhouette scores are low, the clusters are convenience groupings, not distinct farm types, and should not drive decisions on their own.

6. API and testing

Endpoint	Purpose
GET /health	Service status, models loaded, verification code
GET /model-info	Selected models, test metrics, fingerprint, disclaimer
POST /predict	Typed farm record -> yield, risk, probability, cluster, code

Valid request

```
{
 "rainfall_mm": 1000.0,
 "avg_temp_c": 18.0,
 "soil_moisture_pct": 55.0,
 "fertilizer_kg": 120.0,
 "field_size_ha": 1.5,
 "season": "A",
 "cooperative_support": 1
}
```

Response (CLI and API return the same fields)

```
{
 "verification_code": "AI-A2-8180",
 "yield_estimate_tons": 12.3208,
 "high_risk_prediction": 0,
 "high_risk_probability": 0.0246,
 "cluster": 0
}
```

Invalid request and failure behaviour

The request is validated by typed Pydantic fields (`extra='forbid'`, min/max ranges equal to the training ranges). A bad value, unknown field or missing field returns HTTP 422 with a list of problems; missing model files return 503. `predict.py` prints the same problems and exits with code 2. Example: rainfall_mm=2500 gives 'outside allowed range [600.0, 1400.0]'.

Automated tests (tests/test_system.py)

Test	Covers
valid_inference_returns_all_fields	valid inference
missing_field_is_rejected	missing field
wrong_type_is_rejected	wrong type
range_error_is_rejected	range error
wrong_seed_is_rejected / leakage	seed rule, target leakage
reproducibility_same_fingerprint_and_metrics	two runs, same SHA-256 and metrics
dataset_has_required_problems	rows, missing values, duplicates
api_endpoints	/health, /model-info, /predict 200 and 422

Run `pytest -q` and attach the output as evidence/..._TEST_LOG.pdf.

7. Model card and reflection

Intended use. Classroom prototype that helps a field officer estimate harvest, flag possible high-risk farms and group similar farms.

Synthetic data. All records are generated from a formula with random noise; no real farm data was used. Metrics describe how well the models recover this formula, not real Musanze farms.

Limitations. Only seven inputs; uniform independent features; no weather extremes, pests, prices or soil type; ranges limited to 600-1400 mm rainfall and 0.2-4 ha fields; the test set has only 53 rows so scores vary; clusters are weak (silhouette 0.19).

Risks. Over-trust in a score; a wrong 'low risk' result delaying support; unfair treatment of small farms if used outside its purpose; applying the model to farms unlike the training data.

Prohibited use. Deciding access to loans, insurance, subsidies or public services; replacing an agronomist; use on real farmers' data without validation and consent.

Human review. Every output must be reviewed by a qualified person who can override it.

With real data I would collect cooperative records with consent, add soil type, variety and pest information, validate on a later season (time-based split), check performance by season and farm size, calibrate probabilities, and choose the risk threshold with the cooperative.

Classroom decision-support prototype trained on synthetic data. It does not replace an agronomist and must not decide access to finance, insurance or public services. Human review is required.